Participant number:

Information sheet – study Multilingualism and Ageing

Participant Information Sheet **Multilingualism and Ageing (MultiAge)**
It is important to be aware of the information that is provided by the Department of Psychology about the general terms and conditions that apply with respect to the processing of personal data. Please consult our departmental privacy notice (link: https://wiki.york.ac.uk/display/PsySharedDocs/Key+information+about+GDPR)
1) Background The University of York, project leader Dr Angela de Bruin, Dr Farah Nazir, and Dr Zlatomira Ilchovska, would like to invite you to take part in the following research project. Before agreeing to take part, please read this information sheet carefully and let us know if anything is unclear or you would like further information. If you have any questions or want to discuss any aspect of the study, please contact Angela de Bruin and the project team (multiage-project@york.ac.uk or phone: 07435017504) before completing the consent form.

2) What is the purpose of the study?

In this multilingualism and ageing study (called MultiAge), we want to examine how younger and older bilingual adults use their languages and switch between them. We are focusing on people who can speak English and Mirpuri-Pahari-Pothwari and/or Panjabi. We want to understand better the differences between older and younger adults, and see if language changes throughout a person's life. In the future, we also would like to understand how language changes in people with cognitive impairment. Before we do that, we first want to understand better how language changes with ageing more generally. Your data might be used in the future as a "control group" of healthy younger or older adults, to compare with people who have cognitive impairment.

In this part of the MultiAge study, we are comparing younger and older bilingual adults on various language tasks. As MultiAge is a study that follows language throughout time, we would also like to invite you again in a few years to take part in another study session.

Please only take part in the current session if you are also willing to consider taking part again in a few years from now.

What will I be asked to do?

In the current study session, we will ask you to complete a few tasks, namely:

- Picture-naming tasks. In these tasks, you will see simple pictures of objects (e.g., an apple) and would have to name the object out loud. We will ask you to sometimes use English, and sometimes Mirpuri-Pahari-Pothwari, and at other times you will be able to choose which language(s) you want to use. If you speak Urdu, we will also invite you to complete an additional picture-naming task in English, Mirpuri-Pahari-Pothwari and Urdu. However, this final task is not compulsory for all participants and you can still take part in the study if you do not speak Urdu.
- Scene-description task. In this task, you will see pictures of simple scenes (e.g., a girl eating pasta in a kitchen). We will ask you to describe the

scene verbally in one sentence. We will again tell you at each part of the task which language(s) to use.

- A short interview and story telling task, in which we will ask you simple questions (for example, what you did the past weekend) and ask you to tell short stories. We use this task to see how fluent you are in each language and to look at how you produce short stories in both languages.
- A short task in which you watch a video showing you different rooms in someone's home. Afterwards we will ask you some questions about the video.
- A short cognitive task, to examine your overall cognitive functioning. Cognition includes our abilities to think, remember, reason and take decisions in our daily lives. The task includes different short exercises in which we will ask you to memorise short pieces of information, repeat simple sentences, or count shapes.
- A questionnaire that asks about your background and how you use your languages in your daily life.

You will not receive individual feedback on any of the tasks.

You will be audio recorded in the language production tasks (please see our additional consent form on the next pages), so that we can process the data after you do the study.

Where and when can I do the study?

The study will take about 2.5 hours to complete and we can take as many breaks as you like. If you prefer, you can complete the questionnaire by yourself, at a later point after the end of the session. Testing can take place in a location of your choice, for example your home, a local community centre, or in our Psychology department at the University of York.

What do I need on the day?

You do not need to bring anything or prepare anything for the study. Most of the tasks will be completed with the help of a computer, which our researchers will set up for you. You will not have to use the computer yourself, apart from occasionally pressing a button to indicate your response. We will instruct you thoroughly and guide you at every step about this.

3) Why have I been invited to take part?

You have been invited to take part because you speak English and Mirpuri-Pahari-Pothwari and/or Panjabi. We kindly ask you to only take part if you are aged between 18 and 30 years old or between 62 and 80 years old.

Please ONLY take part if:

Language:

- You speak English and Mirpuri-Pahari-Pothwari and/or Panjabi, at least at the level of being able to have a conversation in each language.
- You use both languages at least sometimes in your daily life (i.e., you do not ONLY use English OR Mirpuri-Pahari-Pothwari in your daily life).

Please do NOT take part if:

Health:

- You have been (self-)diagnosed with a neurological or psychiatric disorder, such as major depressive disorder or ADHD.
- You have been (self-)diagnosed with a cognitive impairment or have had a major head trauma that caused a loss of consciousness.
- You have vision difficulties that would prevent you from seeing and naming pictures on a screen, even with glasses or contact lenses. You can take

part if you wear glasses or contact lenses, though – as long as you are able to see the pictures and text on the screen clearly enough to understand them.

- You have hearing difficulties that hinder your daily-life communication and that would make it difficult to follow our spoken instructions. You can take part if you wear a hearing aid, though – as long as you are able to understand and participate in the conversation with the researchers.
- You have (self-)diagnosed language or reading disorders, such as dyslexia or aphasia.
- You are currently taking medication that could affect your alertness and/or concentration levels.

4) Do I have to take part?

No, participation is optional. If you do decide to take part, you will be given a copy of this information sheet. You will be asked to complete a participant consent form. If you change your mind at any point during the study, you will be able to withdraw your participation without having to provide a reason. You can simply tell the research leader to stop.

After study completion, you can contact us to withdraw your research data (your responses in the study) until 31 March 2027. At the start of the study, we will provide you with an anonymous code. If you would like to remove your research data after study completion, please contact Angela de Bruin (angela.debruin@york.ac.uk) with your unique code. If your contact details remain in our database (i.e., if you have not previously indicated that you no longer want to be contacted about this study), we can also find your ID number for you based on your name.

This study is part of the longer MultiAge study, and we would like to ask you to take part again in the future, in approximately three or four years from now. In case we receive further funding, we may also invite you to take part in other future study parts, whose sessions may be scheduled after 2029. If we do not receive additional funding after 2029, we will remove all contact details by

January 2033. You are, of course, free to say no to any future study invitations. You can do this by responding to our invitation. If you do not provide any response to our invitation, we will send two more reminders and stop contacting you if we do not receive a further response. If you change your mind after the first part of the study, you can email Dr Angela de Bruin, in order to be removed from our study invitation database. This is possible at any point.

5) Will you share my data with 3rd parties?

We are collecting the following types of personal data (data that can identify who you are):

- Your name, email address, and home address for payment processing purposes (see the "participant compensation" section).

- Your name, email address, and phone number to invite you to take part in the future sessions of the study. We will also use these contacts to send you an occasional update about the study (for example, to send you a summary of the study findings).

- Audio recordings during the language production tasks.

- Personal background and language background questions. These questions do not ask for much detail, but some questions (e.g., region you are living in) could indirectly identify who you are.

All personal data (data that can identify who you are) will only be accessible to the project team at the University of York. Your personal data will be stored on the project leader's (Angela de Bruin) secure University of York lab folder and/or in a Google Drive spreadsheet. The University of York has an agreement with Google including assurances around data protection. These files can only be accessed by the MultiAge project team. Your name, email address, and home address need to be shared with our departmental administration team so that

they can process the payment. We will not share these personal data with anyone else.

Our research group is committed to open research practices, which includes sharing research data through public repositories such as the Open Science Framework. This way, the data you provide during the study tasks will be shared publicly with others. We will never share any data that can identify who you are to other people (e.g., we would never share your name or date of birth). Technically speaking, the data we share are pseudonymous rather than anonymous. This is because we still have a link between the data and your name (through a pseudonymous code), to make sure we can connect your responses across the testing points. This link is only known to the research team at York. When we share the data publicly, nobody outside our research team would be able to link those data to you. Where we share questionnaire data, we do not share answers that can identify who you are. For example, we would not share individual responses to the question about the region you are living in, or would recode the data in a way that cannot identify individuals. For the language production tasks, we do not share the audio recordings publicly. We only share publicly the information we derive from these recordings (for example, which words you produced and how fast you produced them). For the longer production tasks (the short story-telling task and scene-description task), we might share transcripts of the sentences that you produced. If you provide any information that could identify you, we would edit this before sharing.
The data we share can be accessed by other people who might use them for other research questions that are not known to us. The research team and others might therefore use these data for secondary research and/or teaching purposes.

Other researchers we are working with might be interested in using the audio recordings from the language production tasks for further analyses. We might therefore share not just your pseudonymous research responses but also these recordings with them, if their research questions align with the questions we are asking here. If we share any data with these collaborators, they will first need to complete a data sharing agreement. As part of the data sharing agreement, they will need to confirm they will save the data in a secure and confidential manner

and will not share your recordings with others. We might also use your data as part of the “control group” of healthy younger or older adults in future studies looking at language changes in people with cognitive impairment.

Please only take part in the study if you agree with this.

6) Will you transfer my data internationally?

We will not share any data internationally that can allow others (outside the research team) to identify who you are. We will share research data as explained above.

7) Will I be identified in any research outputs?

No

8) Recompense

To recompense you for taking part in this research, you will receive a £25 Amazon voucher.

If you prefer to receive the participant compensation in cash (for example, if you do not have an email address or no opportunity to use an Amazon voucher), please let us know when we schedule the appointment with you, or at the start of the study session.

If you have to travel to the testing location by car or public transport (for example, if you prefer to take part in the study at our lab, at the University of York, even though you do not live in York), we will cover your travel time and travel expenses (an additional £10 per 30 minutes).

In order to pay participants, we are required to ask for their home address (in addition to their name and email address) to meet HMRC requirements. To be

paid for this study, we therefore ask you to provide us with this information. If you do not want to share this, we kindly ask you not to take part.

Your data (name, email address, and home address) will be shared with the administration team in the Department of Psychology so that they can process the payment. In line with HMRC retention requirements, these data will be kept for seven years. Your data will not be shared with others and will not be used for any other purposes. They will also be saved separately from the data collected in this study. For more information about processing of personal data, please see departmental privacy notice (link: https://wiki.york.ac.uk/display/PsySharedDocs/Key+information+about+GDPR)

Further information about payments:

Please be aware that if you are paid more than £1000 over the course of a year, you will need to complete a Self-Assessment form for HMRC. Being paid for involvement in research is likely to have implications for you whether you are currently employed, unemployed, receiving state benefits or retired. The payment you receive will be treated as earnings by HMRC. There are a number of ways in which receiving payment for involvement may affect your current financial situation. You can find further information here. (link: https://docs.google.com/document/d/1zEduiXdGXYTK-xG2k9EUNvBQ6jo8c8SLyMprQvsbooI/edit?usp=sharing)

9) Questions or concerns

If you have any questions about this participant information sheet or concerns about how your data is being processed, please contact Angela de Bruin in the first instance. This project received ethics approval from the Ethics Committee in the Department of Psychology at the University of York. If you have any questions you would like to ask the ethics committee, please contact psyc529@york.ac.uk. If you are still dissatisfied, please contact the University's Data Protection Officer at dataprotection@york.ac.uk.

Contact Details:

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multiage-project@york.ac.uk

phone: 07435017504)

Participant number:

RECORDING CONSENT FORM Multilingualism and ageing

1) **What kind of recordings will be made?**

We will record your audio (voice) during the language production tasks. This includes the picture naming task, scene description task, short interview/story-telling task, and a task as part of the cognitive functioning assessment in which you have to name as many objects in a category as possible. We will only record you for the duration of these tasks. For the interview task, please do not provide any information about yourself that you do not want us to know.

2) **Who will have access to the raw recordings?**

The recordings will only be accessible to the project team at the University of York, led by Angela de Bruin. They might be shared with collaborators, as explained in Q4.

3) **How will the recordings be made and stored?**

The recordings will be made automatically by the computer we are using to show the tasks on. They will be saved on the project supervisor's password protected lab drive. The recordings will be deleted at the end of the project (in 2035).

4) **Will the raw recordings be shared with others outside of the research team?**

Other researchers we are working with might be interested in using the recordings for further analyses. We might therefore share not just your pseudonymous research responses but also these recordings with them, if their research questions align with the questions we are asking here. If we share any data with these collaborators, they will first need to complete a data sharing agreement. As part of the data sharing agreement, they will need to confirm they will save the data in a secure and confidential manner and will not share your recordings with others. The data will be transferred to them in a secure manner.

5) **Do you intend to produce anonymised versions of the recordings for sharing?**

No. We will use the recordings to measure the things we are interested in (for example, which language you use and when you switch). We will analyse and share these measures, but we will never share any version of the recordings publicly. We might share transcripts in the tasks that ask you to produce sentences (the scene description task and short story-telling task). If you say anything that could identify who you are, we will remove this information from the transcript.